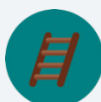


7.1 Factoring and the Distributive Property

Lesson Introduction

 Benchmark	MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system.		 Learning Targets	- I can rewrite a polynomial using the distributive property by factoring out a common factor. - I can determine the greatest common monomial factor.
 Benchmark Coherence	Building On MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.		Working Toward MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real- or complex number system.	
 Essential Questions	- How can you rewrite a polynomial using the distributive property? - How can you determine the greatest common monomial factor?			
 Lesson Narrative	In this lesson, students extend their prior learning about the distributive property to factoring polynomial expressions by a common monomial factor. Students will be introduced to the concept of common monomial factors and the greatest common monomial factor. They will practice factoring polynomials by a common monomial factor with the distributive property. The lesson focuses on rewriting multiple equivalent expressions for a given polynomial.			
 Component Summary	7.1.1 Warm-Up (~5 min.)	Electric Motorcycle Engineer career video (<i>SE p. 357</i>)	7.1.4 Guided Practice (10 min.)	Rewrite polynomials using common factors and the distributive property (<i>SE p. 361</i>)
	7.1.2 Exploration (10-15 min.)	Explore connections between factors and products using the distributive property (<i>SE p. 358</i>)	7.1.5 Your Turn! (5 min.)	Practice rewriting polynomials using common factors and the distributive property (<i>SE p. 361</i>)
	7.1.3 Guided Instruction (10 min.)	Determine the greatest common monomial factor in polynomial expressions (<i>SE p. 360</i>)	7.1.6 Wrap-Up (~5 min.)	Ticket Out the Door rewriting an expression using common monomial factors (<i>SE p. 362</i>)
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> Equivalent <u>Additional Glossary:</u> N/A	(Optional) Warm-Up video	N/A	

 Teacher Tips	Common Misconceptions • Students may make errors when identifying the greatest common monomial factor. • Students may make basic divisibility rule errors when looking for common factors. • Students may try to factor out a variable that not all terms share.	Things to Consider • Students have already been exposed to factoring using the distributive property in either Grade 8 Prealgebra or Grade 7 Accelerated Mathematics, but if students are placed into Algebra 1 without going through the 8th-grade benchmarks, then factoring using the distributive property will be new to them.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect In 7.1.3 Guided Instruction, students receive formal instruction on common monomial factors. This concept may need additional review of divisibility rules. Consider leveraging this routine to help share thinking out loud while walking students through the process.	MA.K12.MTR.4.1: Discussion and Reflection 7.1.2 Exploration has students work together to explore the distributive property in factoring. Students will work with a partner to engage in structured discussions and reflections that lead to the concept of determining common monomial factors in the following activity.
 Differentiate Instruction	7.1.2 Exploration has students work together in a collaborative setting to explore the distributive property in factoring. Consider pairing students together strategically based on their learning styles and prior formative data. Consider creating an anchor chart for the process of determining the greatest common monomial factor of monomial terms. This anchor chart can capture the processes described in 7.1.3 Guided Instruction, with specific reference to the table in Question 6 for examples. The 7.1.6 Wrap-Up for this lesson provides a formative assessment point to determine students' mastery with common monomial factors prior to the next lessons.	
Lesson Notes:		

7.1.1 Warm-Up: Electric Motorcycle Engineer

Component Overview: Students will engage in a review of a quote from an electric motorcycle engineer. This activity is designed to be done independently.



7.1.1 Warm-Up: Electric Motorcycle Engineer

1. Marc Fenigstein says, "Open-ended problems are the problems students will face in the real world ... Forcing students to learn how to function in ambiguity and find their own answers ... and that there isn't necessarily any one right answer, but there might be better or worse answers ... I think that is really valuable. Don't be afraid to find your own path ... You're not going to get it right on the first try, but you can make almost any job your dream job by looking for opportunities to do more, to go beyond ..."

- a. What stands out to you the most in this statement?

Answers will vary. What stands out most to me is how open-ended problems are what we will face in the real world. What stands out to me is the idea that you can make any job your dream job.

- b. Why does that part of the statement stand out to you?

Answers will vary. It stands out to me because those are the problems I avoid in class, but I should embrace them to prepare for the real-world. It stands out to me because it helps me see that I can take any job I'm in and make it amazing.



Display and show the video to students prior to their responding to the prompts.

Consider highlighting any exemplar responses that connect to MTR Standards. For example, highlight responses like "I like how in this class we talk more about how we got to the right answer than the actual answer itself," as this connects to multiple representations (MA.K12.MTR.2.1) and flexibility in fluency (MA.K12.MTR.3.1).

7.1.2 Exploration: The Distributive Property in Factoring

Component Overview: Students will work together with a partner or small group to explore the distributive property in factoring.



7.1.2 Exploration: The Distributive Property in Factoring

- Multiply $9(10 + 6)$ using the distributive property.
 $(9)(10) = 90$
 $(9)(6) = 54$
 $90 + 54 = 144$
- Discuss with your partner how $9(10 + 6)$ shows that 9 and 16 are factors of 144.
Answers will vary. The product of 9 and 16 is 144. Therefore, those two would be factors of 144.
- Discuss with your partner how multiplying $8(287)$ can be done more easily using the distributive property.
 Explain or show your work.
Answers will vary. Taking the value of 287 and breaking it into the sum of other terms can make multiplying easier. $8(287)$ can be rewritten as $8(280 + 7)$ to become $2,240 + 56$, which is 2,296.
- Rewrite the expression $-7(2x - 5)$ using the distributive property.
 $(-7)(2x) = -14x$
 $(-7)(-5) = 35$
 $-14x + 35$
- Complete the table.

Expression	Factors	Product
$9(10 + 6)$	9 and 16	144
$-7(2x - 5)$	-7 and $2x - 5$	$-14x + 35$
$(8 - 4x)10$	$8 - 4x$ and 10	$80 - 40x$
$-12(7 - 3x)$	-12 and $7 - 3x$	$-84 + 36x$
$2(6x^2 + 8x)$	2 and $6x^2 + 8x$	$12x^2 + 16x$
$4x(3x + 4)$	$4x$ and $3x + 4$	$12x^2 + 16x$
$(6x + 8) \cdot 2x$	$6x + 8$ and $2x$	$12x^2 + 16x$



This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group during 7.1.3 Guided Instruction.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.



How did you break the number 287 up to apply the distributive property more easily?

Why do you think it's possible that you can break it into different sums and get the same answer when you distribute?

7.1.2 Exploration: The Distributive Property in Factoring (continued)

6. What do you notice about the product of each of the expressions $2(6x^2 + 8x)$, $4x(3x + 4)$, and $(6x + 8) \cdot 2x$?
All of the products are the same.

Expressions are **equivalent** if they have the same value no matter what number is substituted in for each of the variables.

7. The expression $14x^2 + 28x^3$ is the product of each of the following expressions.

$$2(7x^2 + 14x^3) \quad x^2(14 + 28x) \quad 7x^2(2 + 4x)$$

List the factors for each expression.

Factors of $2(7x^2 + 14x^3)$		Factors of $x^2(14 + 28x)$		Factors of $7x^2(2 + 4x)$	
2	$7x^2 + 14x^3$	x^2	$14 + 28x$	$7x^2$	$2 + 4x$

8. Write two more expressions that are equivalent to $14x^2 + 28x^3$.
*Answers will vary. $14(x^2 + 2x^3)$ and $2x(7x + 14x^2)$.
 $14x(x + 2x^2)$ and $7(2x^2 + 4x^3)$.*
9. Compare your expressions with your partner's expression. Write the expressions that you and your partner do not have in common.
Answers will vary.
10. Complete the table for the factors of $14x^2 + 28x^3$ using two other expressions you created.

Monomial Factor	Binomial Factor
2	$(7x^2 + 14x^3)$
x^2	$(14 + 28x)$
$7x^2$	$(2 + 4x)$
2x	$(7x + 14x^2)$
$14x^2$	$(1 + 2x)$



Consider keeping a copy of this table on a note card with the correct answers. This will help when you are circulating the room to quickly check students' answers to see who may need additional support reviewing these concepts.



Consider creating a visual representation of the word "equivalent." This can include multiple examples with justification for why those examples meet the conditions of equivalency.



Consider having students compare their work with an additional student partner pair to look for additional similarities and differences with the equivalent expressions in question 9.

7.1.3 Guided Instruction: Common Monomial Factors

Component Overview: In this Guided Instruction, students will receive formal instruction on determining common monomial factors, including the greatest common monomial factor, for a given expression.



7.1.3 Guided Instruction: Common Monomial Factors

The table in the exploration shows there are several different ways to rewrite the expression $14x^2 + 28x^3$.

1. What monomial factor is the greatest common monomial factor of $14x^2 + 28x^3$?

$14x^2$

2. Discuss with your partner how to determine if a monomial factor is the greatest common monomial factor of a given polynomial. Summarize your discussion.

When the polynomial being factored can no longer be factored further with additional monomial factors, you have found the greatest common monomial factor.

3. Determine the greatest common factor for 24, -36, and -54.

6

4. Determine the greatest common factor for x^7 , x^{10} , and x^{13} .

x^7

5. Determine the greatest common factor for x^3y^3 , x^2y^8 , and xy^4 .

xy^3

6. Work with your partner to complete the table. If an expression does not have one of the greatest common factors listed, write N/A.

Expression	Greatest Common Numeric Factor	Greatest Common Variable Factor	Greatest Common Monomial Factor
$25w^3 - 40w^2 + 60w$	5	w	$5w$
$81g + 108g^2 - 12g^4$	3	g	$3g$
$7m^3n^2 + 56m^2n^3 - 21mn$	7	mn	$7mn$
$12p^2q + 16q^2 - 64p$	4	N/A	4



Students learned to find the greatest common factor of numerical expressions in Grade 6. Guide students through the process of finding common factors and the GCF of monomials in this component.



What is the difference between a common factor and the greatest common variable factor?

How can you tell if the common factor you found is the greatest common variable factor?



If time permits, consider turning question 6 into a matching activity. You can do this by taking the answers and placing them on index cards. Distribute the cards to all students and have them go around the room to find their correct match.

7.1.4 Guided Practice: Factoring With the Distributive Property

Component Overview: Students will engage in structured practice with factoring with the distributive property. Students should work independently or with their partner to apply what they learned in the prior activities.



7.1.4 Guided Practice: Factoring with the Distributive Property

- Which of the following expressions are equivalent to $8a^3 - 98a$? Select all that apply.
 - ☐ $a(8a^2 - 98)$
 - ☐ $2(4a^3 - 49a)$
 - ☐ $2a(4a^2 - 49)$
 - ☐ $2a(4a^3 - 49a)$
 - ☐ $4a(2a^2 - 49a)$
 - ☐ $8a^3(2 - 98a)$
- Write three equivalent expressions for $18x^3 - 9x^2 - 15x$. One of the expressions should include the greatest common monomial factor.
 - $3x(6x^2 - 3x - 5)$
 - $x(18x^2 - 9x - 15)$
 - $3(6x^3 - 3x^2 - 5x)$
- Write an equivalent expression for $2c^3d^3 + 14c^2d + 18cd$ that uses the greatest common monomial factor.
 - $2cd(c^2d^2 + 7c + 9)$



Use observational data from the 7.1.2 Exploration and 7.1.3 Guided Instruction to determine how much guidance is needed to bring together key ideas in this Guided Practice component. If necessary, consider modeling question 1 to address common misconceptions shown in prior components prior to releasing students to work on questions 2 and 3.



Think-Pair-Share

Consider leveraging a Think-Pair-Share strategy for this component. Have students try to complete these questions on their own and then work with a partner to compare their answers to come to consensus on each question prior to reviewing the component as a whole class.



For a challenge, have students compete to see who can come up with the most equivalent expressions for question 2. This can help students build up their factoring skills.

7.1.5 Your Turn!

Component Overview: Students will engage in an independent practice opportunity writing equivalent expressions using factors and the distributive property.



7.1.5 Your Turn!

1. Write three equivalent expressions for $72b^2k^3 - 64b^5k^2 + 32b^8k^6$. One of the expressions should include the greatest common monomial factor.

$$8b^2k^2(9k - 8b^3 + 4b^6k^4)$$

$$b(72bk^3 - 64b^4k^2 + 32b^7k^6)$$

$$2(36b^2k^3 - 32b^5k^2 + 16b^8k^6)$$

2. Which expressions are equivalent to $48r^3 + 18r^2$? Select all that apply.

☐ $3(16r^3 + 6r^2)$

☐ $6r(8r^2 + 3r)$

☐ $2r^2(24r + 9)$

☐ $4(12r^3 + 4r^2)$

☐ $6r^2(8r + 3)$



For this component, consider working with small groups of students who may have been struggling throughout this lesson.



Consider having students leverage an agree or disagree protocol. Display each expression and have students decide whether that expression is equivalent, and then have students vote on whether they agree or disagree with the answer before turning and talking with a partner to discuss their rationale.

7.1.6 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door to demonstrate their levels of understanding of common monomial factors.



7.1.6 Wrap-Up: Ticket Out the Door

- Write three equivalent expressions for $12a^3b^5 - 16a^2b^2 + 20a^3b$. One of the expressions should include the greatest common monomial factor.

$$4a^2b(3ab^4 - 4b + 5a)$$

$$2(6a^3b^5 - 8a^2b^2 + 10a^3b)$$

$$a(12a^2b^5 - 16ab^2 + 20a^2b)$$



Student performance on this component can be used to determine what additional support, if any, may be needed in future lessons. Students may understand how to find common factors but miss the greatest common factor. Students may also factor incorrectly. Use responses from this Wrap-Up to inform differentiation in Lesson 2.